pfi

Python for Fixed Income

COURSE 4 · MACHINE LEARNING

Dated Data and Walk-Forward Validation

WEEK 3 · DAY 1

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Week 3: rows that have an order

Week 1

A row is a bond. No order.

Week 2

A row is a card account. No order.

Week 3, today: a row is a date

2,940 dates of Treasury par yields, 2 Jan 2015 to 2 Oct 2026. Each date sits next to the day before and the day after, which look almost exactly like it.

Target: the same day's 30 Yr yield. Never a later date.

Every model today describes the curve's shape on a date already in the file. Nothing is a forecast.

Source: U.S. Department of the Treasury, Daily Treasury Par Yield Curve Rates (home.treasury.gov), retrieved 3 October 2026; public domain (CC0, Treasury data inventory).

A date's neighbours are near-copies

465 bp

30 Yr range over the file, 0.99 to 5.64 percent

3.0 bp

Median move from one date to the next

0.9989

Correlation of a level with the day before

Daily changes do not repeat

One day's change against the next: correlation -0.0207. Levels sit next to near-copies; changes do not (section 11.7).

Day 1's "rows that are not independent": a test date's near-copy, the day before, sits in the training rows.

The task, and the names it may have

Given the same day's 2 Yr and 10 Yr yields, the 30 Yr yield that goes with them.

Allowed names

fitted 30 Yr yield, same day's 2 Yr and 10 Yr

out-of-sample error inside the file's span

Never called

a forecast, an expectation

fair value, a view on rates, a signal

Evaluation stays inside 2 Jan 2015 to 2 Oct 2026. Nothing is fitted or reported after the last date.

How DecisionTreeRegressor works

Same tree	Week 2's tree, with a number as the target
The question	Scored by how much it lowers the squared error: each side is summarized by the mean of its targets
The leaf	Holds the mean target of the training rows that landed there: that is the output
Left at defaults	It grows until its leaves are nearly pure
What comes out	A number that is always an average of training values, never outside their range

A random split: the tree looks excellent

```
random_splits = list(KFold(n_splits=5, shuffle=True, random_state=SEED).split(X))
```

4.5 bp

Tree, shuffled folds: MAE; R-squared 0.9949

97.1%

Of 588 test dates with a training date one row away (none over 2)

10.8 bp

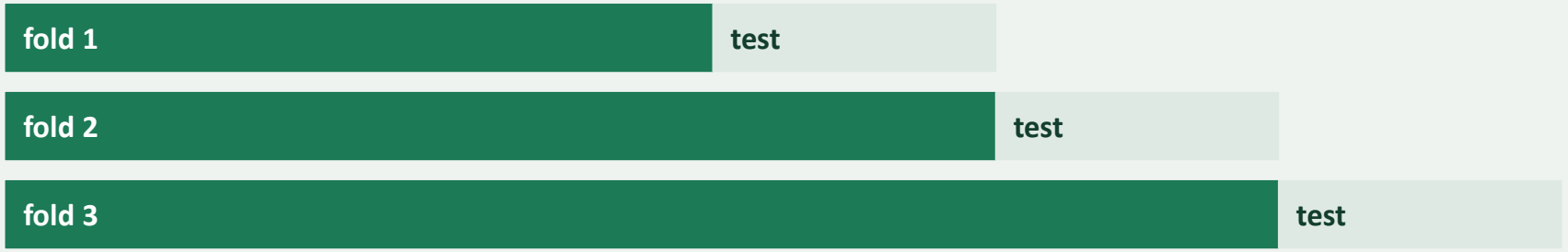
Straight line, the same folds: it cannot copy a neighbour

On shuffled dates the tree finds yesterday and reads off its yield. That measures memory, not the curve's shape.

How walk-forward validation works

Fold 1	Trains on the first stretch of dates, tests on the stretch right after it
Next folds	Add the test stretch to the training rows, then test on the one that follows, to the end of the file
The rule	Every training date is before every test date; nothing is shuffled; each date is tested once
You control	The first training length, the test window, and the gap: here 504, 126, and 0 rows
What comes out	A list of folds, each a pair of row positions, and one error per fold

Walk-forward: train on the past, test on what follows



Setting	Rows	About
min_train	504	two years
test_size	126	six months
gap	0	same-day target

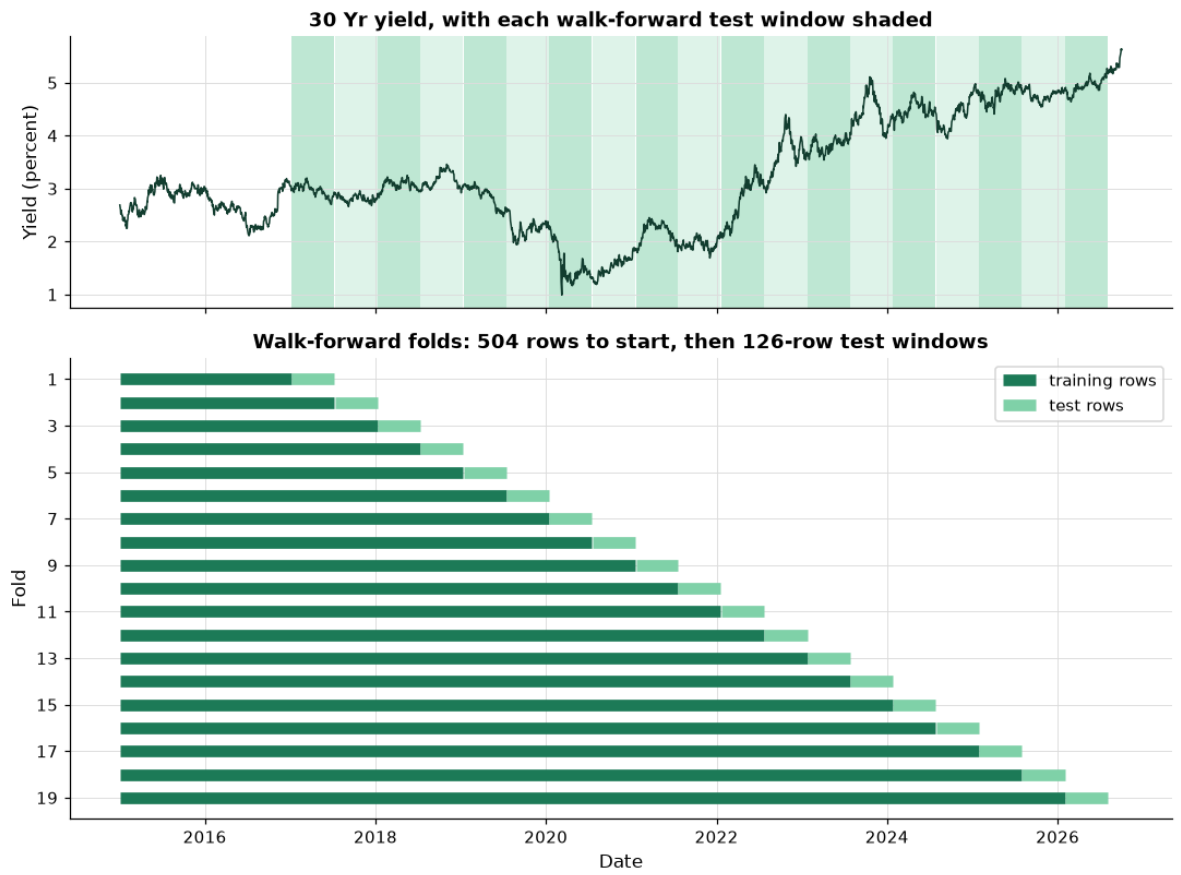
mlkit.walk_forward_splits

19 folds on the Treasury file; the last 42 dates do not fill a window.

mlkit.assert_past_only

Every training date before every test date. 47 tests pass.

Nineteen folds over the file



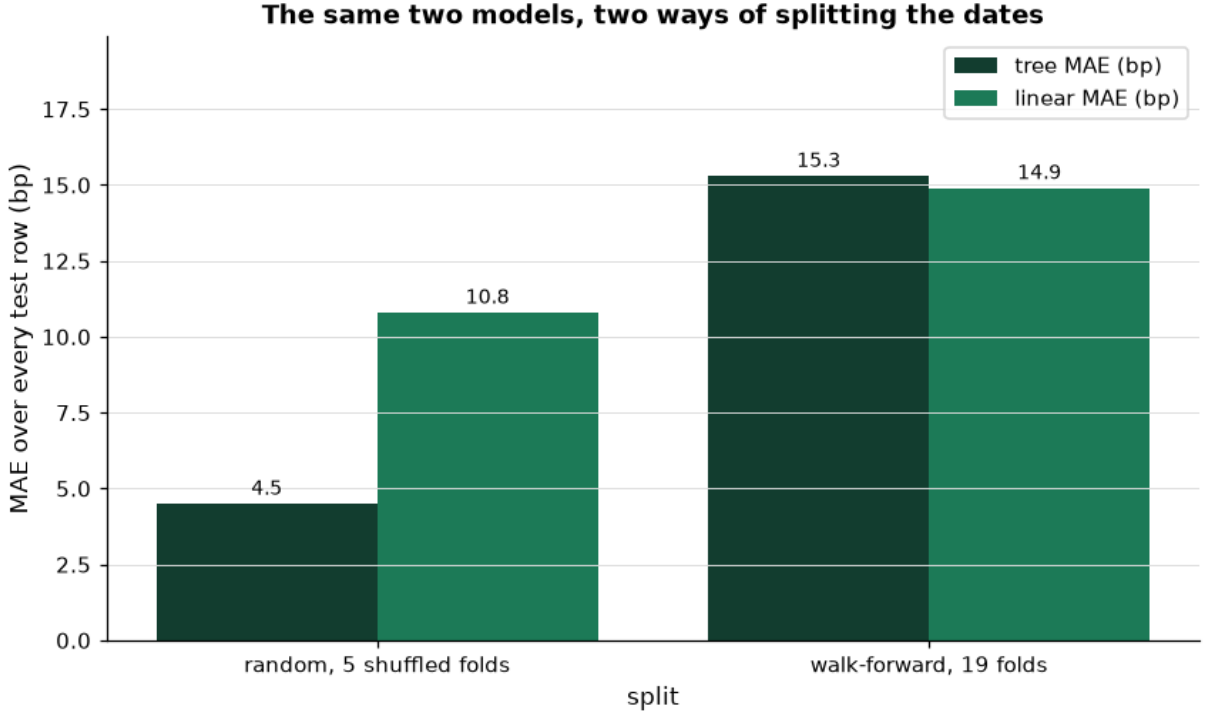
A lane per fold

Dark: training rows, from 2 Jan 2015, growing by 126 rows a fold. Light: the next 126 test rows.

Each date tested once

The test windows tile the file; the last 42 dates do not fill one.

Random against walk-forward



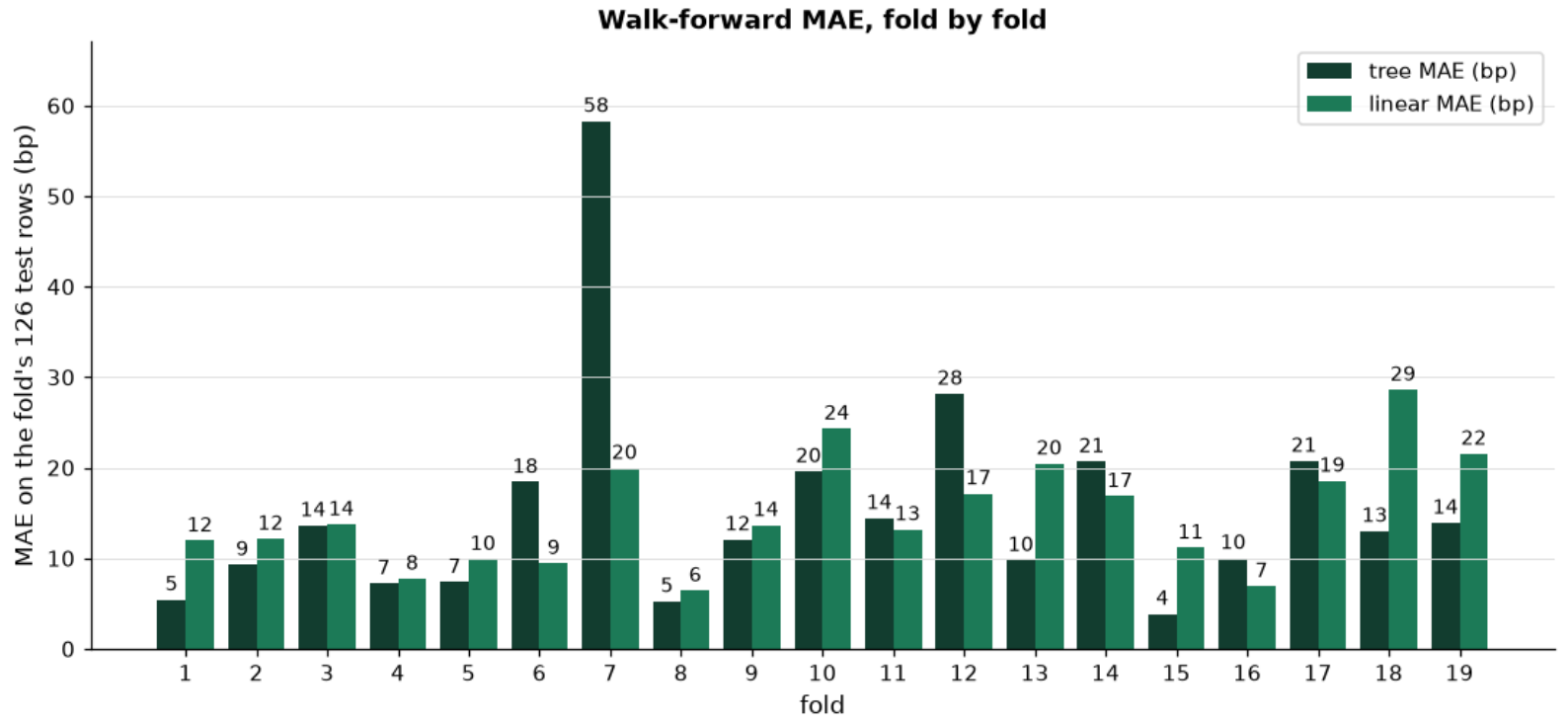
Split	Tree (bp)	Line (bp)
Random, 5 shuffled	4.5	10.8
Walk-forward, 19 folds	15.3	14.9
TimeSeriesSplit(5)	23.0	25.8

The ranking flips

Random: the tree far ahead. In time order: the two are close.

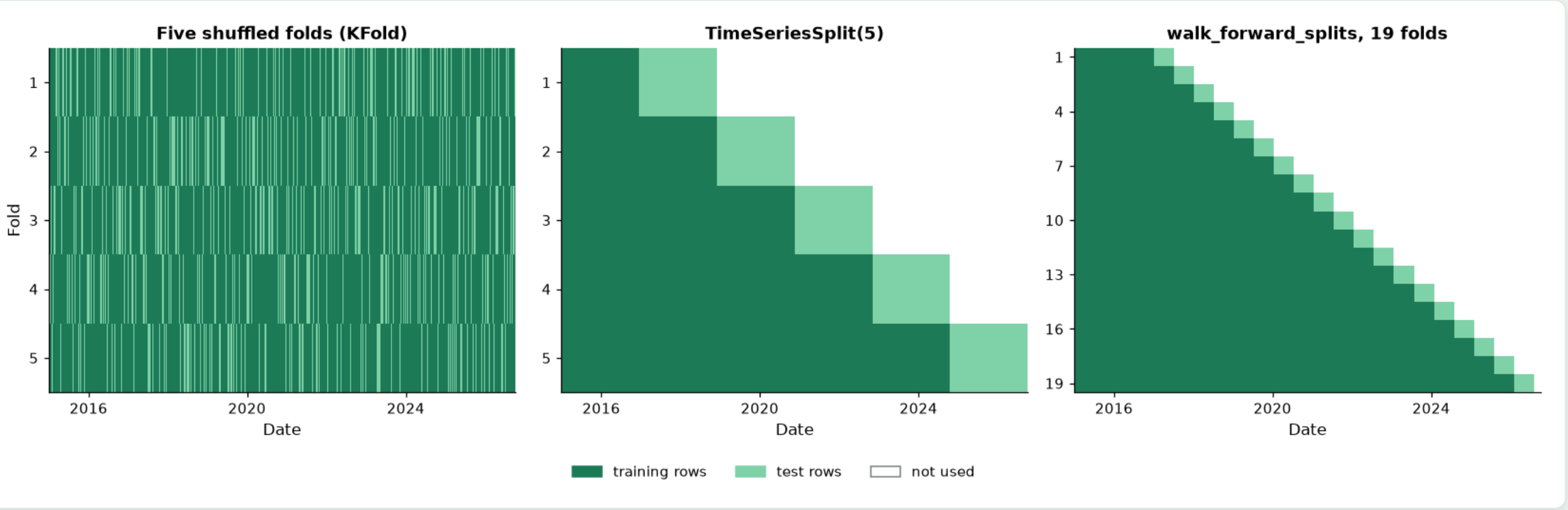
Not forecast errors: out-of-sample errors inside the file's span, on later dates already in the file.

Error, fold by fold



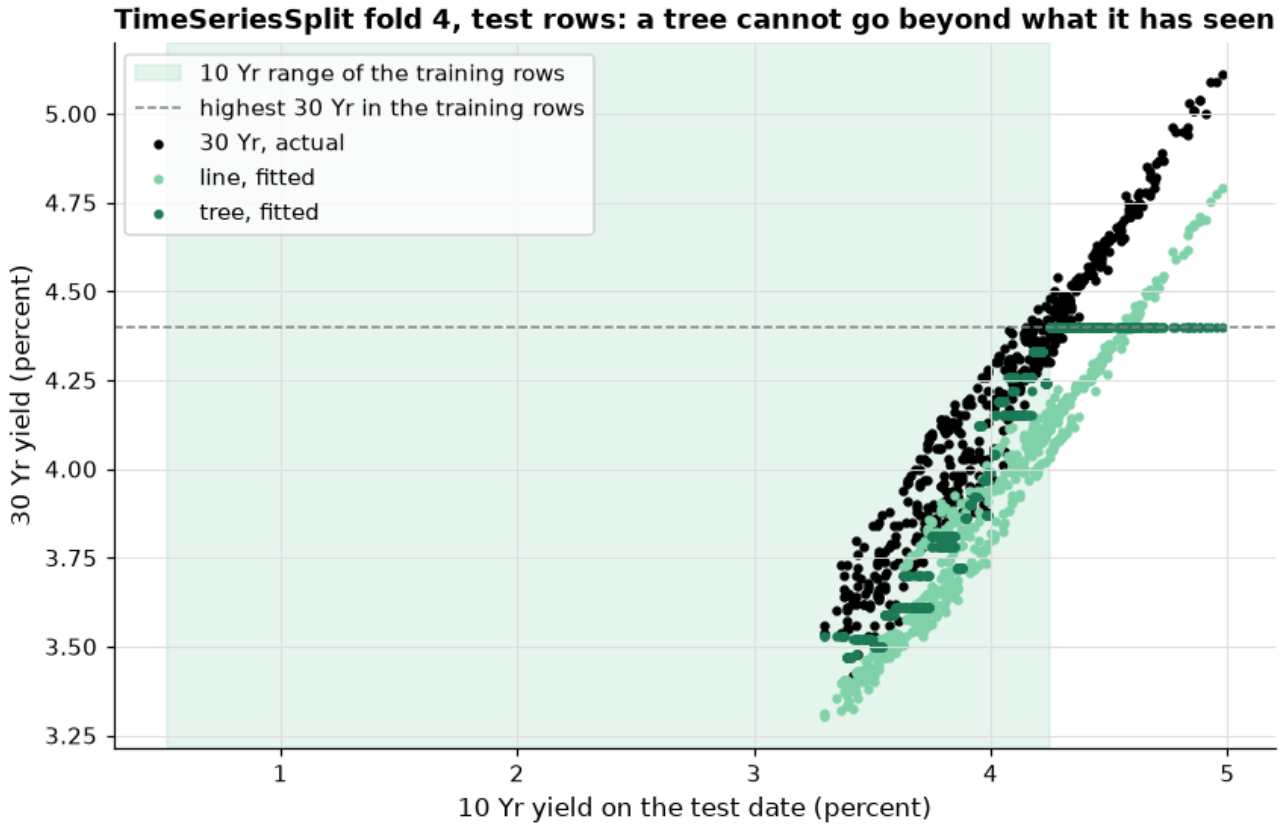
Fold 7, 16 Jan 2020 to 16 Jul 2020: tree 58.3 bp, line 20.0 bp. The average says how far; the bars say when.

Three ways to cut the dates



Tree MAE by fold: shuffled 3.9 to 4.6 bp (spread 0.3, mean 4.5); walk-forward 3.8 to 58.3 bp (spread 12.2, mean 15.3). Report a mean and a spread.

Why: levels the training rows never reached



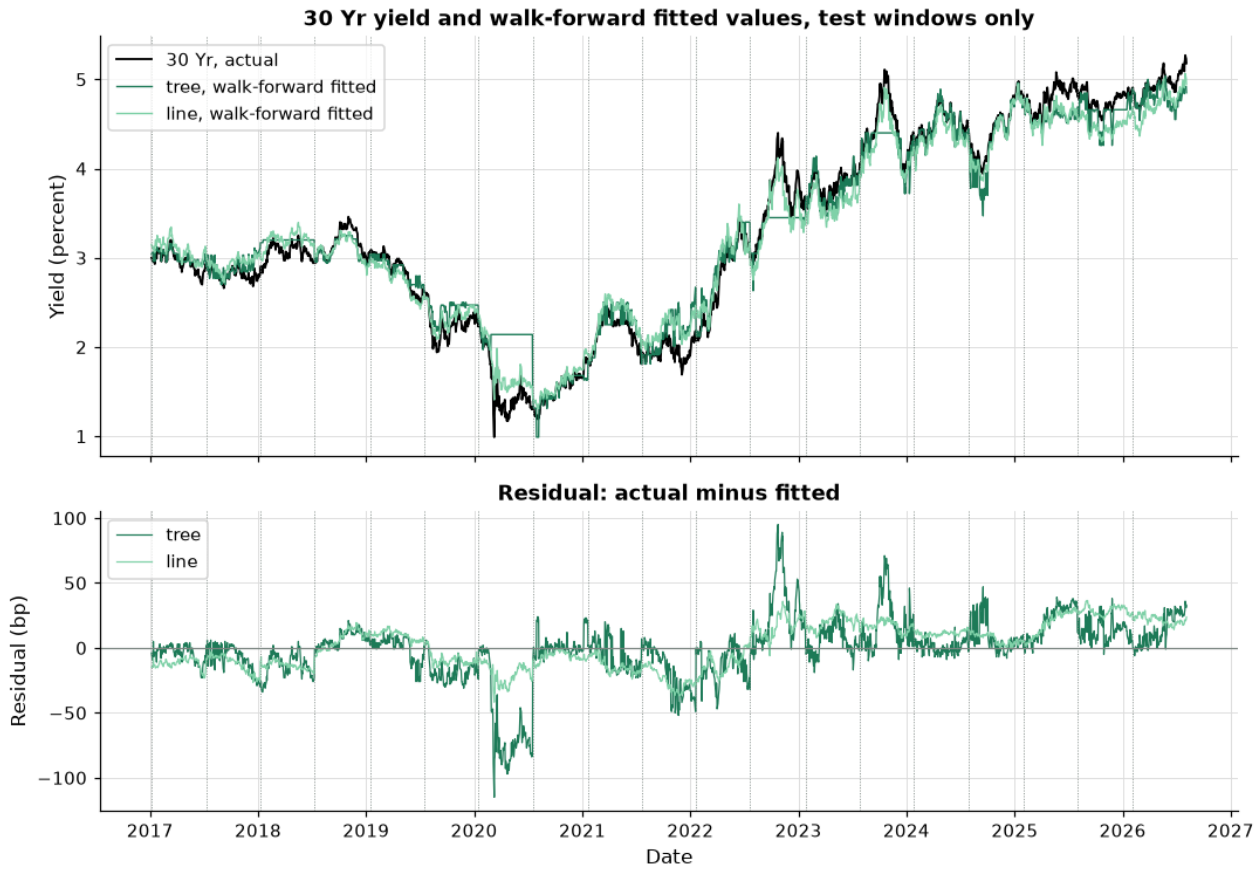
TimeSeriesSplit fold 4

10 Yr: training to 4.25, test to 4.98. 30 Yr test to 5.11; training to 4.40. The tree stops at 4.40; the line reaches 4.79.

Walk-forward fold 7: tree 58.3 bp

16 Jan 2020 to 16 Jul 2020: 30 Yr as low as 0.99 against a training low of 1.94, on 102 of 126 dates. Line: 20.0 bp.

Fitted and residual, test windows only



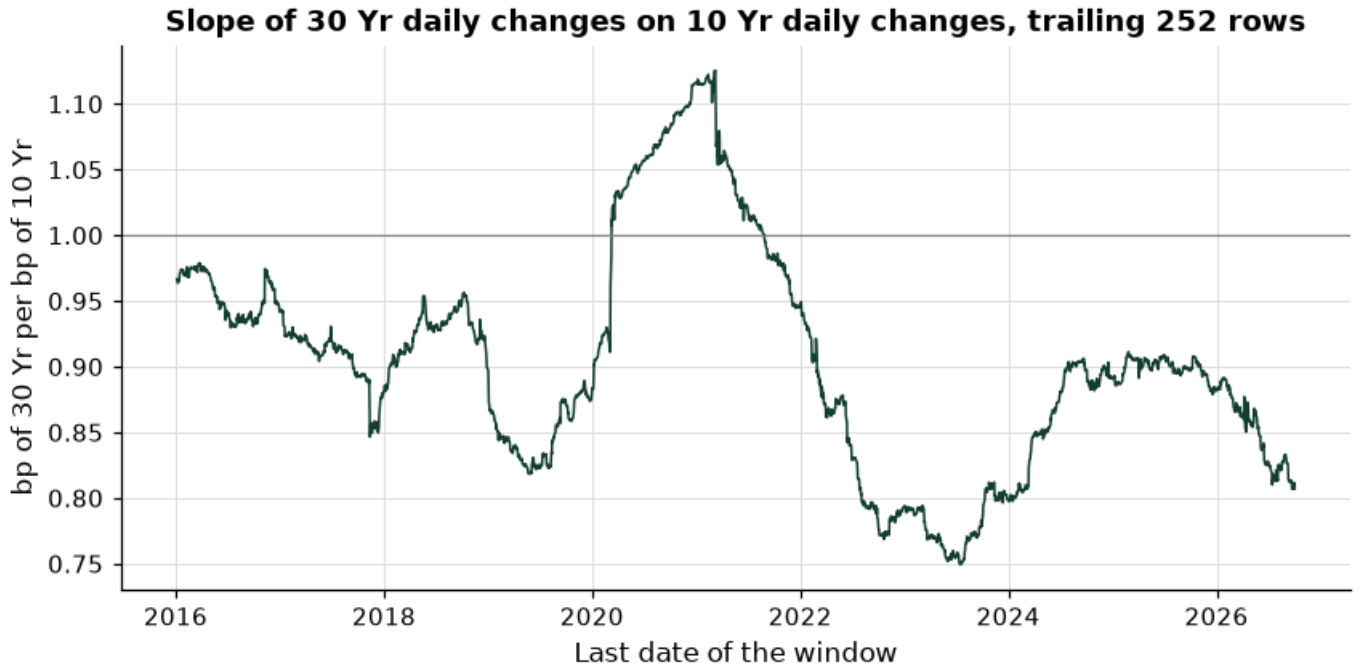
Top: fitted against actual

The tree runs flat outside its training range and jumps when a fold adds rows.

Bottom: residuals (bp)

Largest miss of both on 9 Mar 2020: tree 115.0 bp, line 41.8 bp. The line's residuals change sign 17 times.

A relationship that drifts



0.749

17 Jul 2023

1.125

9 Mar 2021

Last date in the file: 0.807. Each value describes its own trailing window only.

The line's 10 Yr weight

From 0.9004 (fold 6) to 1.1843 (fold 19).

A random split hides drift: its training rows come from every stretch at once.

Ridge and lasso where the features are collinear

30 Yr daily change given the same day's changes in the other ten tenors, in bp.

0.9529

Correlation, 20 Yr and 10 Yr changes

60.58

VIF of the 7 Yr; 6 columns above 10

Walk-forward R-squared	OLS	Ridge	Lasso
Expanding, 19 folds	0.9604	0.9604	0.9600
Rolling 63 rows, 115 folds	0.9464	0.9475	0.9497
10 Yr weight std, rolling	1.6387	1.1273	0.8857

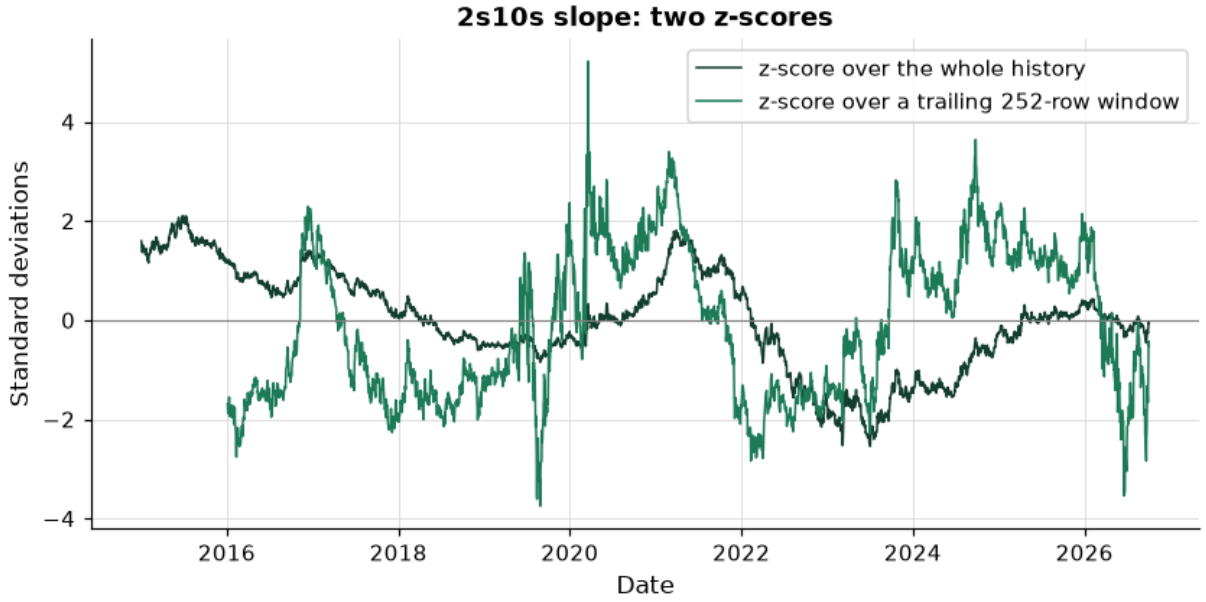
Lasso sets the 6 Mo to zero in the last expanding fold. Alpha is chosen with TimeSeriesSplit inside each fold.

Few rows and near-copy features: shrinkage helps a little. Thousands of rows: almost no change.

How a trailing z-score works

The problem	A whole-history mean or spread uses dates after the one being scored
Trailing window	For each date, the 252 rows ending at that date
The z-score	Standardize the date's value against that window's mean and standard deviation
The test	The value on date t is the same whether the data stops at t or runs on
What comes out	A feature that could be computed live, empty until the window fills

Lookahead in features



z-score on 31 Dec 2019	All dates	Data stops
Whole history	-0.2303	-0.9311
Trailing 252 rows	2.3634	2.3634

1,250 of 1,250 whole-history values change when the data stops. That is lookahead.

Durbin-Watson: 0.01

About 2 for unrelated residuals; near 0 when each copies the last.

Build a feature from the past only: a trailing window, or a step fitted inside the pipeline.

Excel side by side

Excel side by side (tested in Excel)

`=(C3-C2)*100  (F3 down)`	daily change in bp, exact
`=CORREL(G3:G2941,F3:F2941)`	0.9529
`=SLOPE(OFFSET(H254,-251,0,252,1),OFFSET(F254,-251,0,252,1))  (I254 down)`	0.749 to 1.125
`=TREND($E$2:$E$505,$B$2:$C$505,B506:C506)  (J506 down)`	fold 1, fitted
`=AVERAGE(ABS(E506:E631-J506:J631))*100`	fold 1 MAE 11.9 bp
`=STANDARDIZE(K2,AVERAGE($K$2:$K$2941),STDEV.S($K$2:$K$2941))`	whole history -0.2303
`=STANDARDIZE(K253,AVERAGE(K2:K253),STDEV.S(K2:K253))`	trailing 2.3634

The dollar signs are the lookahead: with the data after 31 Dec 2019 cleared, the whole-history z-score moved to -0.9311; the trailing one stayed.

Excel has no walk-forward routine. A fold is two ranges: training rows above, test rows below.

Save the day: git blame

```
Day 11: walk_forward_splits and assert_past_only, with five tests
Week 2: classification on the card file
Week 1: regression on the spread cross-section
```

```
git blame -s --root -L "/def walk_forward_splits/,+1" mlkit.py
```

```
49) def regression_metrics(...)
339) def cv_folds(...)
410) def walk_forward_splits(...)
```

Three lines, three commits: week 1, week 2, and today. The history is rebuilt by week in the work folder.

The AI check

The docstring is the prompt. The code that comes back runs, and has one modelling bug.

1

Docstring

Which rows fit, which rows report, and the units: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Assert what the docstring promises. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: KFold(5, shuffle=True) on dated rows: 4.5 bp, against 23.0 bp in time order.

Code that runs is not the same as code that is right.

What you can do now

- On dated rows, a random split reads the answer from yesterday
- Split by time; prove it with assert_past_only
- Describe a bad fold by its dates and numbers
- Ridge and lasso help when rows are few and features near-copies
- A whole-history feature is lookahead, even in a clean split

Week 3, Day 2 Scaling and K-means: learning without a target